

Nicolas Rohart

Forestry - Design - Robotic Fabrication - Construction

From standing tree to finished structure.

I operate across the entire timber value chain - forest management and harvesting, computational design, robotic shaping of curved timber, and on-site construction - including the digital workflows that connect each stage.

BASE	EXPERIENCE	LOOKING FOR	OPEN TO
Limousin, France	15 years, full chain	R&D partnership	Worldwide

THE CHAIN - FIVE STAGES, ONE HAND THROUGHOUT

01 Forest & Silviculture

Independent forest management, felling and sale, aimed at resilient long-term ecosystems rather than short-term extraction. Developer of "Bois Suivis" - a field-verifiable tracked-timber protocol that goes beyond declared certification schemes.

02 Harvest & Extraction

Felling and skidding done hands-on, in direct feedback with the rest of the chain. Built a custom skidding vehicle from a repurposed 4x4 fire-service chassis - bodywork, crane, structural welding - individually type-approved for road use.

03 Design

Design proposals for custom timber structures from real, round, naturally curved trunks, fitted to each client's brief and site. Rhino 3D for modeling, TwinMotion for renders composited onto photogrammetry scans of the existing sites - near-photorealistic proposals that clients can already picture themselves in.

04 Robotic Shaping

Parametric Grasshopper / KukaPRC workflows generating toolpaths for a 7-axis KUKA robot that follows the tree's actual curvature rather than forcing it into a rectilinear beam. Dedicated 3D scanner: a 3 m trunk becomes a 600,000-point cloud in 3.5 minutes. Carried out at La Manufacture du Châtaignier with mentor Ludovic Mallegol.

05 On-Site Building

Two arrangements available: custom-cut timber pieces to supplied plans, or hands-on involvement through to completion. On-site assembly is carried out with qualified carpenters, coordinating the other trades involved.

NETWORK

- IBOIS (EPFL Lausanne)
- GC2D (Égletons)
- IUT du Limousin (Limoges)

WHAT I'M LOOKING FOR

A research team, studio, or workshop working on wood, robotics, or both - bringing 15 years of hands-on chain experience into a lab or production environment as developer or trainer. Based in France, open to relocating.

Robotic fabrication

Timber engineering

Grasshopper / KukaPRC

Forestry & silviculture

Technical training

PUBLICATION

"The tree dictates the shape: reversing the industrial paradigm in rural areas"

L. Mallegol, N. Rohart, N. Champroy - Construction Robotics (Springer), 2026. Scan-to-trajectory robotic shaping of natural curved timber, developed at Coop Monts et Barrages.

TECHNICAL PROFILE

FORESTRY Forest assessment Silviculture Timber harvesting	DIGITAL FABRICATION Parametric design → toolpath (Rhino / Grasshopper / KukaPRC) Commercial renders (MeshRoom / TwinMotion)	CONSTRUCTION Design Robot operation Site installation
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